

1 Environmental and Ecological Factors (TOR 1)

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Identify relevant ecosystem and climate influences on the stock. Characterize the uncertainty in the relevant sources of data and their link to stock dynamics. Consider findings, as appropriate, in addressing other TORs. Report how the findings were considered under impacted TORs.

The most important environmental driver of the U.S. sea scallop fishery in recent years has been increasing bottom temperatures in the Mid-Atlantic Bight area that has resulted in a reduction in the productivity of scallops in this region. As will be discussed later in this assessment, recruitment in this region has been very low during the last 6 years of the assessment time period (2018-2023), and adult natural mortality has been increasing. This is most evident spatially, with few adult scallops observed in 2023 in the southern and inshore portions of the Mid-Atlantic. Increased adult mortality was first observed in the southernmost surveyed area, off of Virginia, in the early 2000s (Figure 1.1), and has been slowly spreading northward since then. Over half of all U.S. sea scallop landings were from the Mid-Atlantic region between 2000-2019, but only about 5% of landings were from this region in 2023. Further details regarding this phenomenon and an exploratory analysis can be found in the Caracappa et al. (2025) working paper.

There is strong evidence that food supply (mostly phytoplankton, but also detritus and microzooplankton) affects growth and reproductive output of sea scallops (MacDonald and Thompson 1985, 1986ab, MacDonald et al. 1987). Phytoplankton supply declines with increasing depth (Shumway et al. 1987), explaining observed declines in growth, and weight at size (Langton et al. 1987, Barber et al. 1988, Hart and Chute 2009b, Hennen and Hart 2012). For survey data, these effects are taken into account by using shell height to meat or gonad weight relationships that depend on depth. The SAMS model uses area-specific growth and allometric relationships, based on data from surveys and shell growth analysis. For example, the relatively deep water Hudson Canyon area is slower growing with smaller meats at size than average, and the shallow inshore Mid-Atlantic area is faster growing with larger meats at size. The CASA and SYM models, which operate on a region-wide level, cannot explicitly take geographic variations in growth and weights into account. However, the different growth periods built into the CASA model can reflect changes in growth and weights at size over time, possibly due to environmental conditions, and also due to shifts of the distributions of scallops to deeper or shallower areas. Ocean acidification (OA) may affect scallop populations. In particular, there have been several studies demonstrating that OA can reduce the growth and survival of larval scallops (Talmage and Gobler 2009, White et al. 2013, Andersen et al. 2013), and swimming ability (Schalkhauser et al. 2013). Pousse et al. (2023) showed that OA reduces feeding rates and growth of juvenile sea scallops. Similarly, Lagos et al. (2023) showed that OA reduced growth and shell thickness in the Peruvian scallop *Argopecten purpuratus*. Cooley et al. (2015) and Rheuban et al. (2018) used a forward projection model similar to a non-spatial version of SAMS together with forecasts

of future ocean warming and acidification. They predicted that OA will reduce sea scallop yields in the second half of this century, due to reduced growth and larval survival.

Predation can affect scallop distribution and mortality. In particular, the sea star *Astropecten americanus* appears to be reducing or excluding sea scallops from the deeper water of the Mid-Atlantic (Hart 2006, Shank et al. 2012). This sea star consumes a wide variety of small invertebrates, including early post-settlement sea scallops (Franz and Worley 1982, Adebola et al. 2022). *A. americanus* appear to be limited by cold water temperatures, and for that reason, are only common in areas where winter minimum bottom temperatures remain above 5°C (Franz et al. 1981). *A. americanus* densities have been monitored in dredge surveys since 2000, and there is evidence that their distribution is expanded northward and inshore, likely due to warming temperatures (NEFSC 2018). Elevated levels of the sea star *Asterias vulgaris* and concomitantly increased numbers of sea scallop “clappers”, an indicator of natural mortality, were observed on surveys of the northern edge of Georges Bank in 2024, in both U.S. and Canadian waters.

Cancer spp. crabs (*Cancer irroratus* and *C. borealis*) are important predators on juvenile sea scallops (shell heights < 90 mm, Elner and Jamison 1979, Nadeau et al. 2009), and may be agents of density dependence. Wong et al. (2005) seeded juvenile scallops in experimental plots at densities of 1, 6, or 69 m². Scallop density in the high-density sites declined markedly due to both predation, primarily by *Cancer* spp. crabs, and dispersal, resulting in final densities of about 1 m² regardless of treatment. Predation rates of *Cancer* crabs on juvenile sea scallops were greater when scallops are more common than alternative prey species, and increase with increasing scallop density (Barbeau et al. 1998, Wong and Barbeau 2005). This is accounted for in this assessment by directly estimating juvenile natural mortality by year.

The invasive colonial ascidian *Didemnum vexillum* has been observed in portions of Georges Bank and Gulf of Maine. This tunicate can rapidly spread over gravel/cobble substrate, often completely covering it. Scallop larvae cannot settle on *D. vexillum* (Morris et al. 2009), so this tunicate turns preferred settlement substrate into unavailable habitat for juvenile scallops. Consistent with this, Habcam survey data indicate that sea scallops are less common in areas dominated by *D. vexillum* (Figure 1.2, Kaplan et al. 2017). Additionally, this tunicate is more common in fished areas than in an adjacent closed area. This may be due to fishing activity dispersing the tunicates by both direct transport and fragmentation (Morris and Carman 2012). *D. vexillum* appears to be restructuring the benthic invertebrate community in areas that it dominates (Kaplan et al. 2018). Fortunately, *D. vexillum* has been observed to date only in relatively small portions of Georges Bank, mainly on the northern edge near the Hague line border with Canada.

Off-colored “gray meats” have been most commonly observed on Georges Bank, and in particular in Closed Area I (Levesque et al. 2016). They have been associated with infections with Apicomplexa protists (Inglis et al. 2016). However, Siemann et al. (2019) suggested that gray meats are more of a symptom of poor condition, and can occur due to multiple causes. Gray meats due to Apicomplexa infections may have contributed to the increased natural mortality observed in the Georges Bank closed areas during 2011-13. The variable natural mortality included in the Georges Bank Closed Areas model is one method to account for this.

Since 2015, a substantial portion of the scallops in the Mid-Atlantic have been infected with larvae of the nematode *Sulcascaris sulcata*, particularly in the southern portion of the range (Rudders et al. 2023; Figures 1.3, 1.4). The larval stage of the nematode infects scallops and creates brown or orange lesions in the scallop meat, caused by the immune response in the

scallops. The adult nematode lives in the gastrointestinal track of sea turtles, primarily loggerheads, who acquire the nematodes by consuming infected scallops or other benthic mollusks. These nematodes are cosmopolitan and are associated with scallop and sea turtle populations around the world (Lichtenfels et al. 1978, 1980; Lester et al. 1980). It is likely that this outbreak of nematodes was due, at least in part, to the large 2013 scallop year class in the Mid-Atlantic, which may have increased the amount of scallops consumed by the turtles. Increases in the loggerhead turtle population off the northeast US may also be a contributing factor. It is unclear whether nematode infections by the nematodes increase the natural mortality of the scallops. Lightly infected scallops do not appear to be seriously negatively affected by the nematodes, but heavy infections may increase the scallops' mortality. However, the nematodes have had a clear effect on the fishery, as heavily infected scallops are difficult to market (even though there is no public health risk).

Sea scallop blister disease occurs when the shell underneath the gonad gets colonized by a fouling organism, most likely *Polydora* spp. polychaete worms. The scallop then deposits shell material (conchiolin) over this area, creating a "blister" (Figure 1.5). Shell blisters are associated with reduced growth, yield (Figure 1.6), fecundity, deteriorated physical condition, mortality, inhibited water filtration due to reduced shell-cavity volume, and weakened and/or deformed shell (Kent, 1979, Lauckner 1983, Diez et al., 2011). Blisters contain anaerobic metabolites like hydrogen sulfide, which can reduce quality and marketability (Handley and Bergquist 1997). Two decades ago, blister disease was limited to the deeper water in the southern half of the Mid-Atlantic, suggesting that it is limited by cold temperatures. However, blister disease is becoming more common throughout the Mid-Atlantic (Figure 1.7), and has also been observed on Georges Bank and the Bay of Fundy, likely due to increasing bottom temperatures.

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Figures

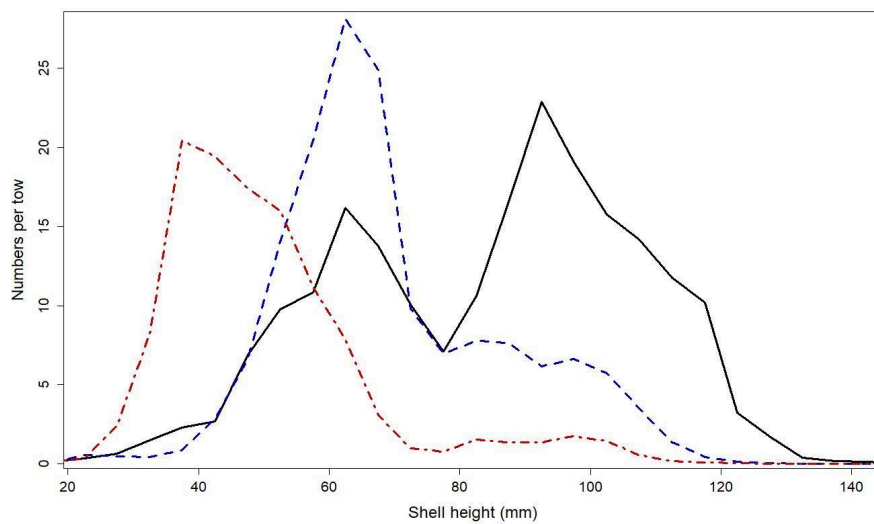


Figure 1.1. Dredge survey shell heights from off of Virginia (between 36.5 and 37.25° N), showing reduced numbers of larger scallops over time, indicating progressively increasing adult natural mortality, from Zang et al. (2022). The black line are shell heights from 1998-2000, the blue dashed line from 2001-2008, and the red dashed-dotted line between 2009 and 2018.

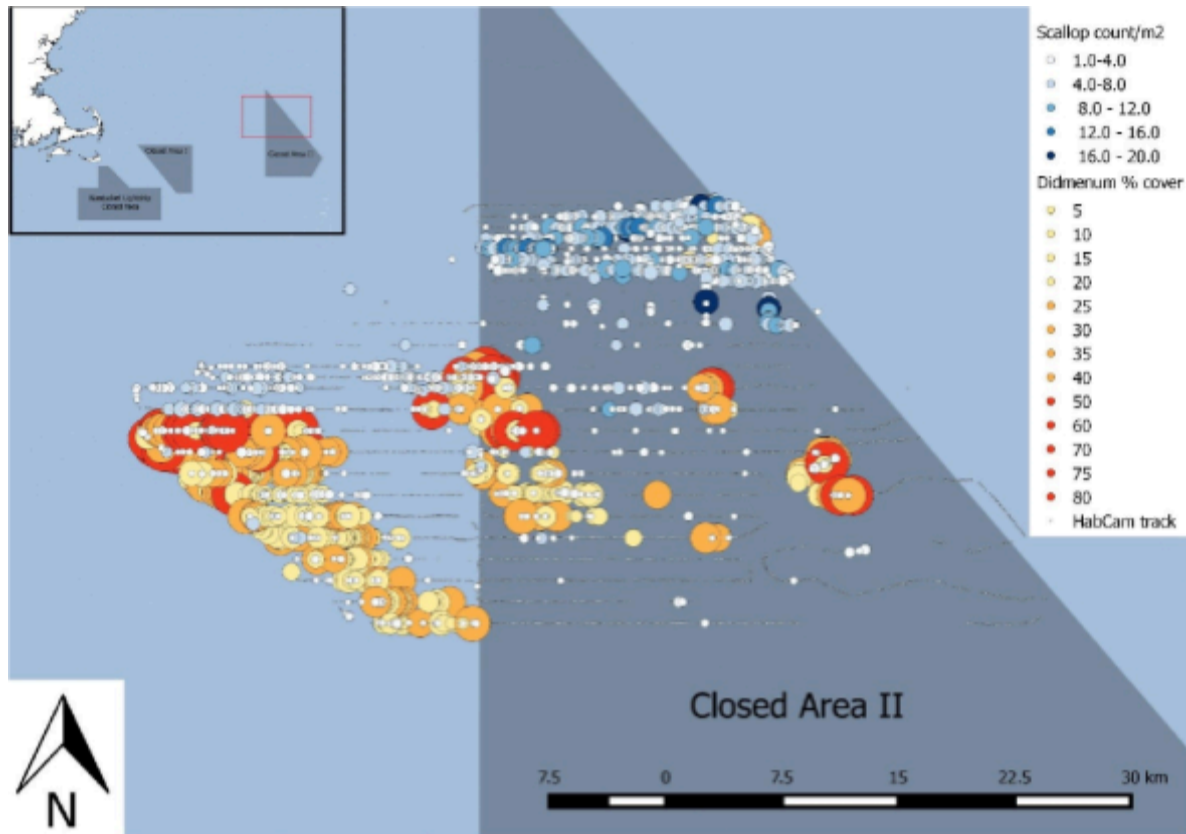


Figure 1.2. Chart of sea scallop (blue dots) and *Didemnum vexillum* (yellow to red dots) observations, from an intensive 2012 Habcam survey of the northern edge of Georges Bank, from Kaplan et al. 2017.



Figure 1.3. A scallop meat, taken from the Elephant Trunk area in the Mid-Atlantic in 2017, heavily infected with nematode larvae, as indicated by the numerous dark orange lesions (Photo by D. Hart).

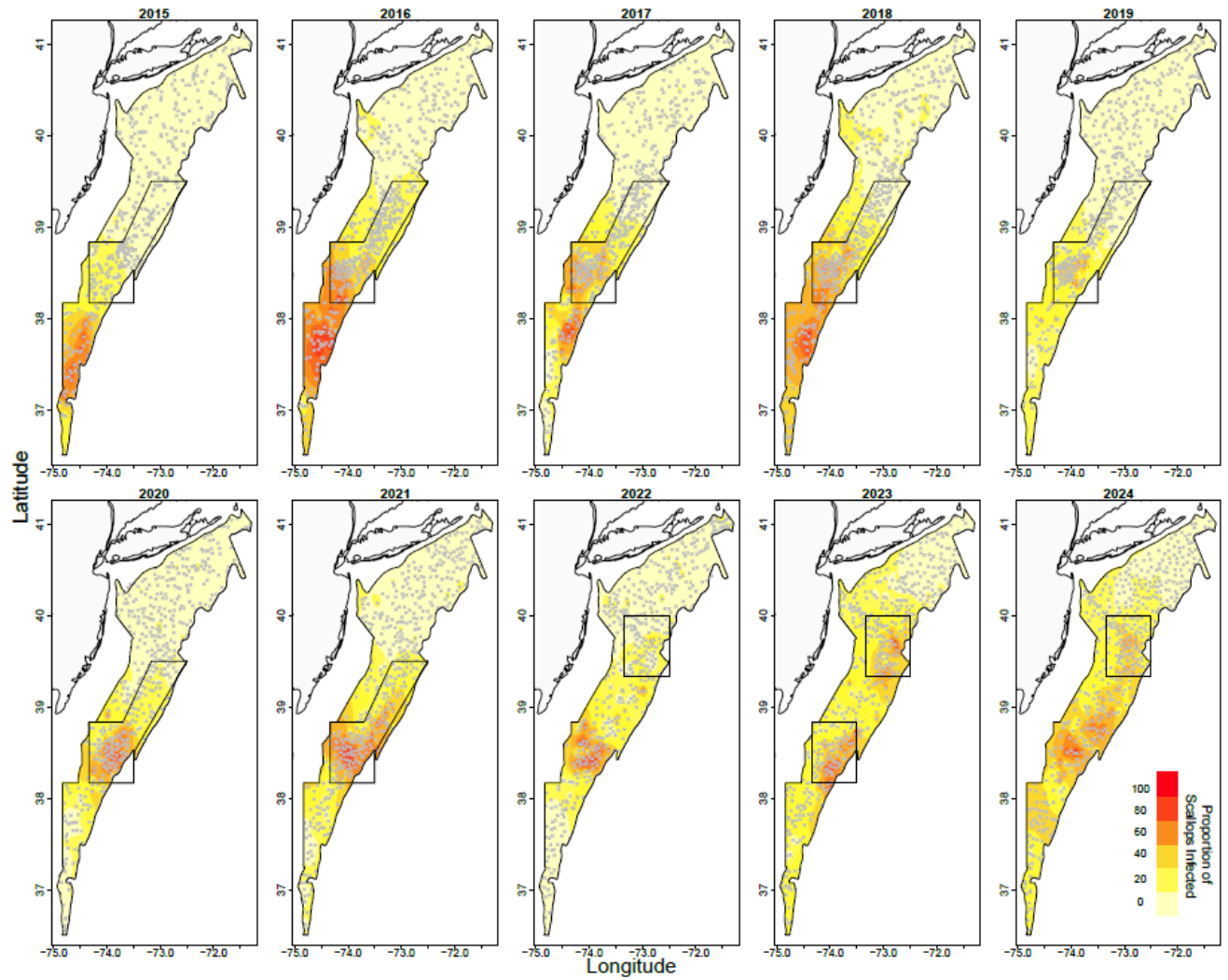


Figure 1.4. Prevalence of larval nematode *Sulcascaris sulcata* infection in scallops in the Mid-Atlantic Bight, from dissection conducted during dredge surveys (Rudders et al. 2023). The colors indicate predicted proportions of scallops infected, and the circles indicate dredge station locations where nematode sampling occurred.

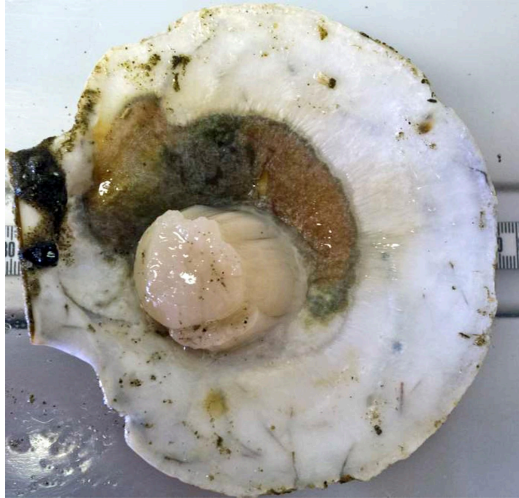


Figure 1.5. Dissected scallop showing a shell blister (Photo by D. Hart).

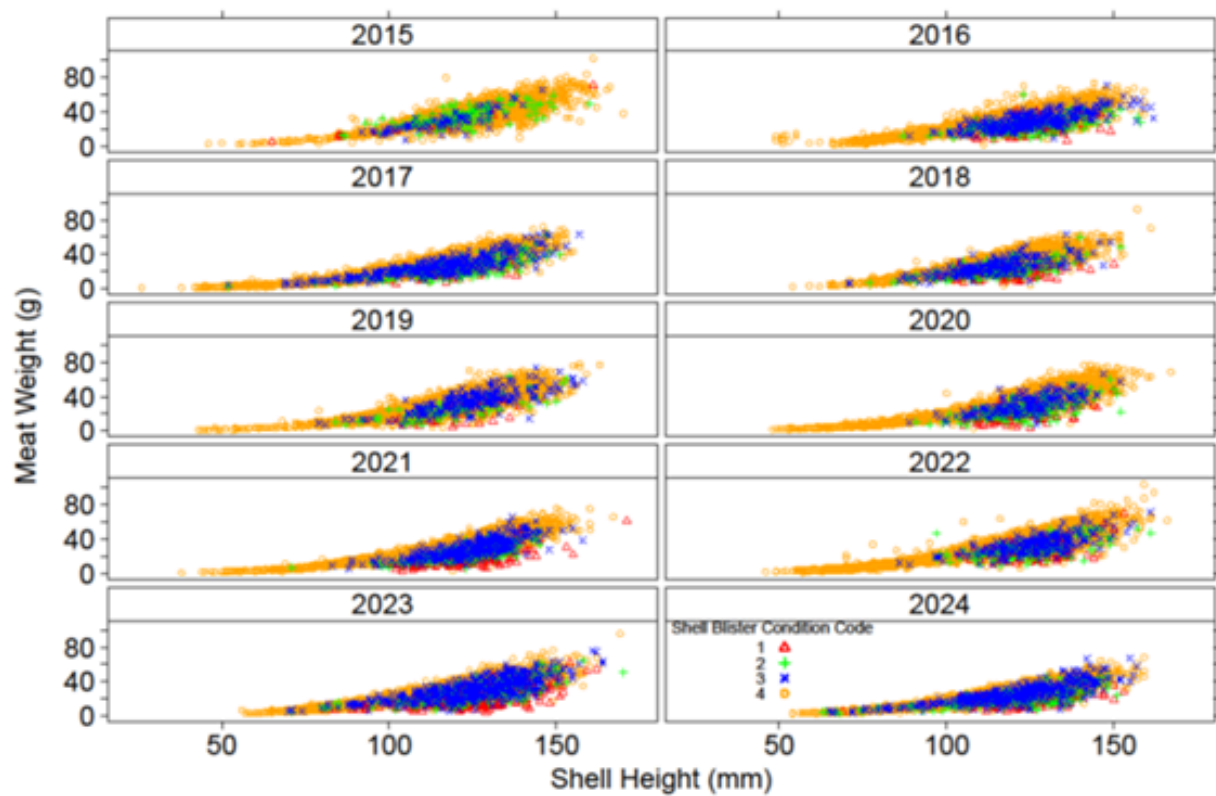


Figure 1.6. Shell height meat weight relationships for the Mid-Atlantic Bight resource area from the VIMS dredge survey for varying degrees of shell blister (2015-2022). Shell disease is assessed with a semi-qualitative score, with condition code 4 (orange symbols) indicating no shell blister and condition code 1 (red) indicating the most advanced stage of shell blister.

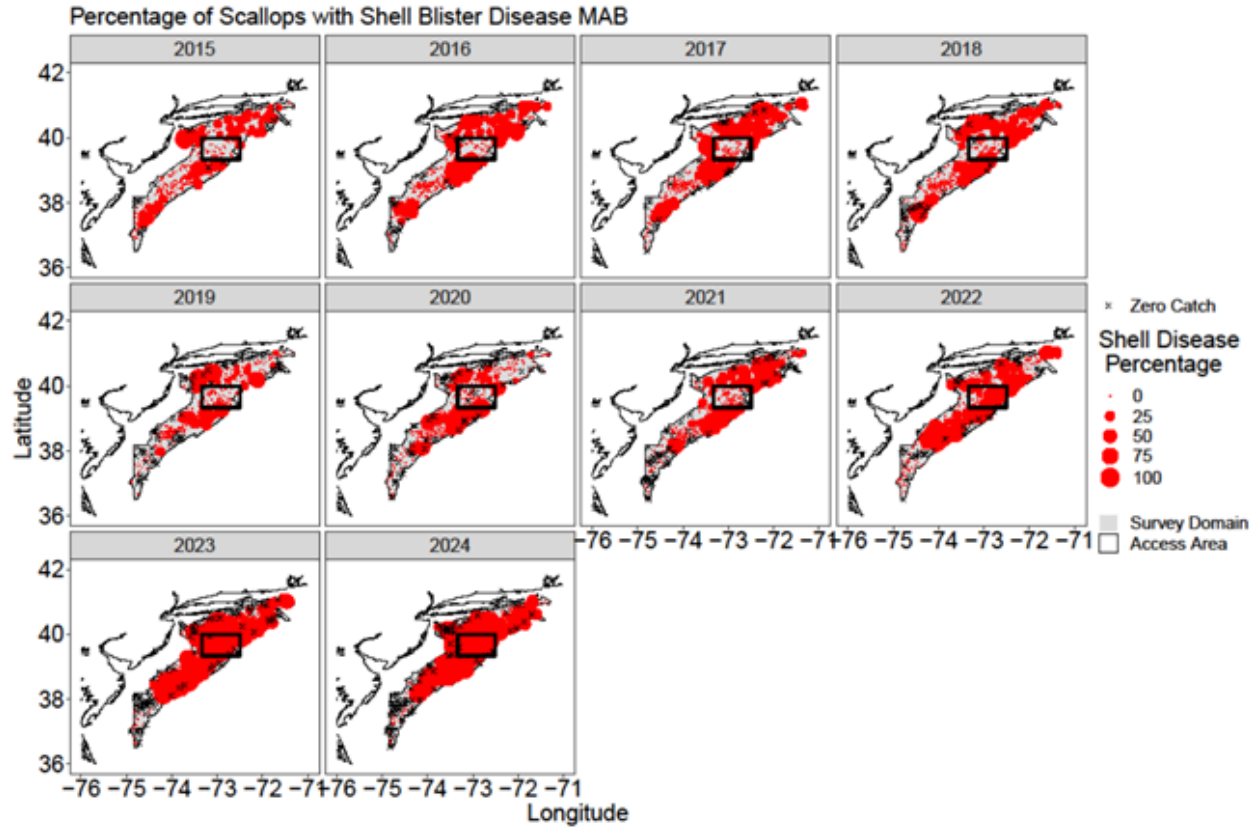


Figure 1.7 Percentage of scallops with shell blister in the Mid-Atlantic for 2015-2024 observed during the VIMS dredge survey. Red circles indicate the percentage of scallops with shell blister, and the black crosses indicate survey stations with no scallop catch.